Understanding the CPU (Central Processing Unit)

1 What is a CPU?

The **Central Processing Unit (CPU)** is the core component of a computer that carries out instructions of programs by performing basic arithmetic, logic, control, and input/output operations.

2 Main Components of a CPU

1. Control Unit (CU)

- Directs the operation of the processor.
- Fetches, decodes, and manages instruction execution.
- Controls data flow between CPU and peripherals.

2. Arithmetic Logic Unit (ALU)

- Handles all arithmetic and logical operations.
- Examples: addition, subtraction, AND, OR, comparison.

3. Registers

- Fast, small memory locations inside the CPU.
- Store temporary data, instructions, and addresses.
- Types include:
  - **Accumulator (ACC)** – Stores intermediate results.
  - **Program Counter (PC)** – Holds address of next instruction.
  - **Instruction Register (IR)** – Holds the current instruction being decoded.
  - **General Purpose Registers (R0, R1, ...)** – Hold operands and results.

3 Basic CPU Operations: The Instruction Cycle

The instruction cycle describes how the CPU executes an instruction:

1. **Fetch** – Get the instruction from memory.
2. **Decode** – Interpret what the instruction means.
3. **Execute** – Perform the operation.
4. **Store (Write-back)** – Save the result, if any.

4 CPU Performance Factors

Several factors determine CPU efficiency:

- **Clock Speed (GHz)** – Higher speed = faster instruction processing.
- **Instruction Set Architecture (ISA)** – Determines available operations.
- **Pipelining** – Allows simultaneous execution of instruction stages.
- **Cache Memory** – High-speed memory close to CPU for frequently used data.
- **Multi-core Architecture** – Multiple cores increase multitasking performance.

5 Fun Fact

Modern CPUs can have over **10 billion transistors**, enabling powerful multi-tasking and parallelism across multiple cores.

6 Instruction Cycle: Step-by-Step

1. Fetch

- **Program Counter (PC)** provides the address of the next instruction.
- Instruction is fetched from memory into the **Instruction Register (IR)**.
- PC is incremented to point to the following instruction.

2. Decode

- The **Control Unit** interprets the instruction in IR.
- Divides the instruction into:
  - **Opcode** – Type of operation (e.g., ADD, LOAD).
  - **Operands** – Data or addresses involved.

3. Execute

- The ALU or relevant execution unit performs the operation.

4. Memory Access (if needed)

- If the instruction requires data from memory, the CPU accesses it at this step.

5. Write-back

- Result of execution is stored back into a register or memory location.

7 The Continuous Cycle

After one instruction completes, the CPU repeats the cycle — beginning with the next instruction pointed to by the PC. This forms the infinite loop called the **Instruction Cycle**.

8 Instruction Cycle Example: `ADD R1, R2, R3`

Assume: Instruction format is `R1 = R2 + R3`